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Are Language Models Actually Useful for Time Series Forecasting? (Appendix)

A Limitations

Here, we discuss the limitations of our paper.

1. We evaluate the ability of LLMs using time-series forecasting. However, to get a better picture of how LLMs can work with time-series, this ability should be evaluated across other downstream tasks as well, such as time-series classification and question-answering.
2. Our evaluation is limited to only time-series datasets, *i.e.*, sequences with even time-intervals. However, there also exists a large fraction of data in the form of non-uniform series, such as payment records, online purchases, etc. Understanding the ability of LLMs in forecasting non-uniform sequences is also necessary to verify our claim on the usefulness of LLMs for time-series data.

B Broader Societal Impact

One of the major impacts our study will have is on the influx of models that use LLMs for modeling time-series. Our results will help researchers to not simply follow the trend of using LLMs in all applications, but to check their usability in detail. Specifically, these findings will help them determine if the LLM component is necessary and if the computational costs are reasonable for the specific setting. In addition to the research community, our findings on the better performance of smaller and simpler models will help develop scalable models that are easy to understand, interpret, and can be deployed cheaply in real-world applications. While we agree that a majority of our results are experimental and limited to selected datasets, we feel that these results will also help researchers narrow down their search space for better models in time-series forecasting, and not simply neglect the simpler models.

C License

All our contributions will be released under the MIT License.

D Additional Experimental Details

D.1 System Configuration

We train and evaluate each reference method and each architecture modification using the same device. For Time-LLM [14], applying LLaMA-7B [33], we use NVIDIA A100 GPU with 80GB memory. For other methods [49, 21], applying GPT-2 [28], we use NVIDIA RTX A6000 GPU with 48GB memory. Though an analysis of memory footprint is beyond scope of this research, we note that training the baselines in the absence of LLM can be done within smaller GPUs as well.

D.2 Baseline Hyper-Parameter Details

When reproducing the reference methods, we used the original repository’s hyper-parameters and model structures. In the ablation study, due to the smaller model parameters, we adjusted the learning rate or increased the batch size in some cases. All other training details remained identical with the reference methods. The specific training details and hyper-parameters are provided in the code and run scripts of the repository². Note that the training process and hyper-parameters for the simple methods can also be accessed via this link.

D.3 Details of Encoder Exploration and Simple Methods

To investigate the source of LLM method performance, we conducted further research on encoders. We used various encoders to encode time series data, followed by a linear layer to project the time

²<https://github.com/BennyTMT/LLMsForTimeSeries>

series embeddings to the forecast. The encoder structure combining patching and attention is shown in Figure 4. The key difference between PAttn and PatchTST [26] is the absence of position embedding and the Feed Forward in PatchTST [26], or, more specifically, replacing the Transformer Encoder with a simple single-layer Attention structure. It could be a simple yet effective method to help evaluate the trade-off between cost and performance for newly proposed methods.

In addition, we propose three different neural models (i) “LTrsf”: which performs better on larger datasets, uses CALF’s [21] encoder without cross-modal attention. This could be a potential source of CALF’s performance; (ii) “D-LTrsf”; and (iii) “D-PAttn”: both of them decompose the time series into three sub-sequences and forecast each using the above two methods respectively, and then linearly combine the results for the final forecast. Across our results in Table 18, we note that even simpler models significantly outperform LLM-based time-series models. In detail, the LLM-based models, all combined we able to appear 33 times as the best and the second-best performer. However, “PAttn” was outperforming them by appearing 34 times as the best and the second-best performer. In addition, the comparison between “PAttn”, “LTrsf” and other state-of-the-art methods is shown in Table 19.

E Additional Experiments

Here we present the results of additional experiments that also highlight the ability of LLMs in modeling time-series data.

E.1 Confidence Intervals for Forecasting

Since LLMs and deep learning models in general are probabilistic in nature, their predictions can vary across different runs and different random initializations. Thus, we report the confidence intervals (CIs) for all MAE and MSE predictions made by our baseline models. We report the CIs for MAE prediction by Time-LLM, CALF, OneFitsAll in Tables 9, 11, and 13, for MSE predictions in Tables 10, 12, and 14, respectively. Across all results, we note that the range of variation is quite small, and these intervals do not affect our observations. To illustrate the subtle differences more clearly, we present the visualized results in Figure 5 and Figure 6.

Confidence Intervals for other Datasets. To evaluate the generality of the ablations in the paper, we introduce five additional datasets that have not been studied by the reference methods [49, 14, 21]. The above datasets are used in many time series forecasting studies [29, 9, 38, 1, 45]. The prediction lengths for the “Exchange Rate” are "96, 192, 336, 720", as in [45, 17]. The prediction lengths for the other four datasets are 30, 48, 56, and 12, respectively, following the settings in Chronos [1]. As shown in Tables 15, 16, and 17, the forecasting performance on the five new datasets, using the three methods [49, 14, 21] we referenced, still demonstrates that language models are unnecessary for forecasting tasks.

E.2 Complete Results

Here we provide the results for all methods and datasets that we were unable to add to the main paper.

Inference Times. All results regarding “inference time” and forecast performance are shown in Figure 7 and Figure 8 respectively.

Randomized Parameters and Random-Shuffling of Inputs. The results of the randomized parameters are shown in Table 20. The remaining results for shuffled input are shown in Table 21, 22, 23 and 24.